

Report Exam Marks Dataset

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1 Introduction

The dataset that is going to be analyzed in this report contains the exam marks from a Probability and Inference exam taken by 150 students. The exam consists of two types of questions, labeled as A and B, with four questions in each group. The A-type questions have a maximum score of 40 points each, whereas the B-type questions have a maximum score of 60 points each. The total exam score sums to 100 points.

The dataset is structured as a matrix where columns represent individual exam questions, and rows correspond to students, with each cell containing the score obtained by a student on a given question. Additionally, a row labeled “max” is included, indicating the maximum possible score for each question. This structure allows for a detailed analysis of student performance and the relationships between different question types.

2 Data visualization

Once we have standardized the variables we can compare the marks of the A questions and the ones of the B questions. We can observe that the values of the A questions are higher than the other ones, since we have most of the values between -2 and 2 and those of questions B between -3 and 1.5.

This result can also be seen also in Figure 2, where the boxplots of the standardized scores for A question and B questions are displayed.

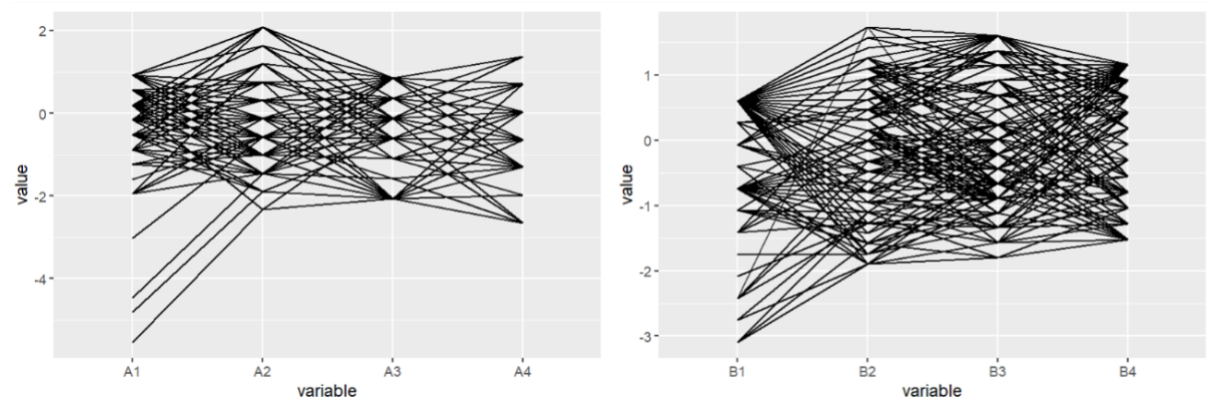


Figure 1: Parallel coordinates plots A vs B marks

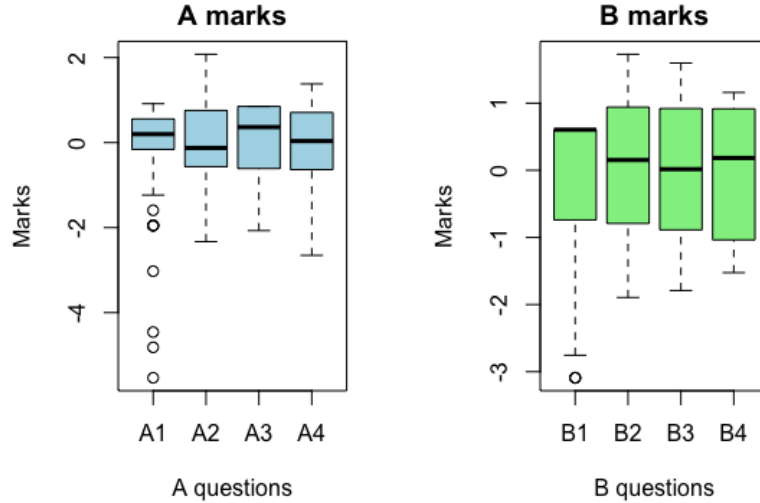


Figure 2: Boxplots of standardized scores

A questions show a slightly lower median and less variability than the B questions, since the boxes in the second plot are wider than the ones in the first plot. A1 presents some outliers indicating lower scores, that could explain the lower median. Overall the interquartile ranges are similar across questions within each group.

3 Statistical Analyses

3.1 Cluster Analysis

In this section we applied cluster analysis, to specify how similar each of the exercises is, and to see if we can group them. The cluster analysis is done on the scaled data, so the comparability should be ensured. As a distance function, we used correlation between variables. The linkage methods used were complete linkage and average linkage.

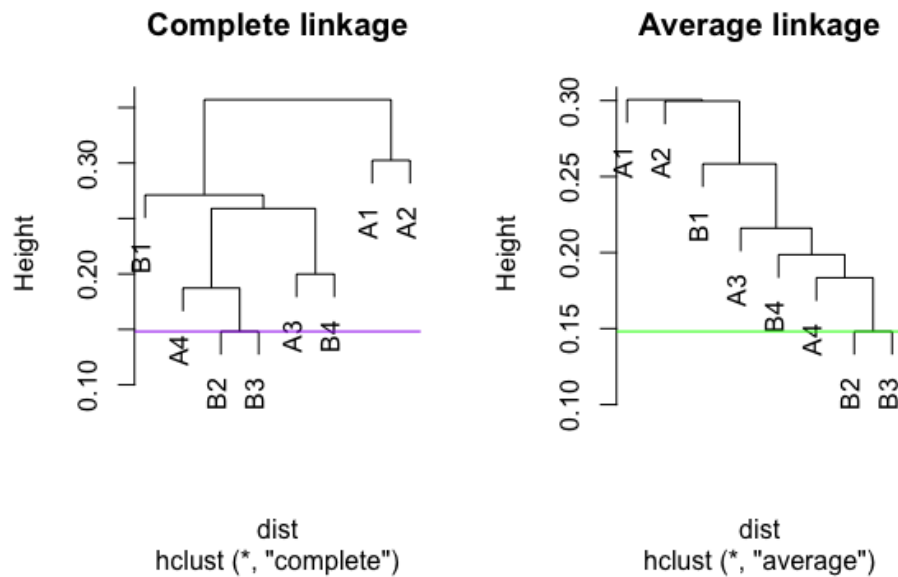


Figure 3: Cluser Analysis Results

The dendrograms above visualize how the questions group together based on similarity. We see that the

largest increase in height (dissimilarity) occurs at the first step of the merging process, suggesting we should keep 7 clusters. This indicates that the exam questions exhibit significant differences, making clear groupings less straightforward. However, B2 and B3 are highly similar, as they are merged first, while A1 and A2 stand out as separate from the others (even if they are merged later). The remaining questions form a larger group.

The conclusion according to both methods is that the exercises B2 and B3 are the only one similar enough to be in single group; this will be seen in more detail in the next paragraphs.

3.2 Principal Component Analysis (PCA)

PCA was performed to identify the linear combinations of the observed variables that capture as much variability in the data as possible. Figure 4 shows high positive correlations among all the exam questions; therefore, we aim at studying how these high correlations may be interpreted thanks to PCA.

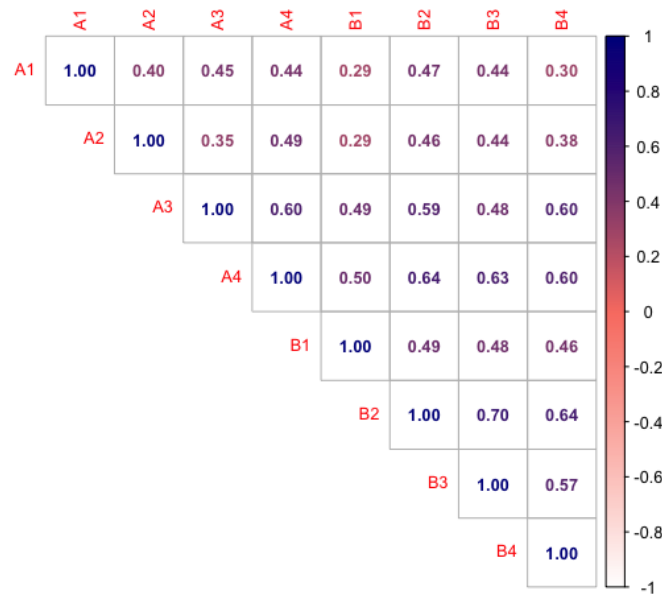


Figure 4: Correlation Plot

We used the `prcomp` function (which relies on singular value decomposition) to perform the analysis. A scree plot was generated to help decide the number of principal components to retain.

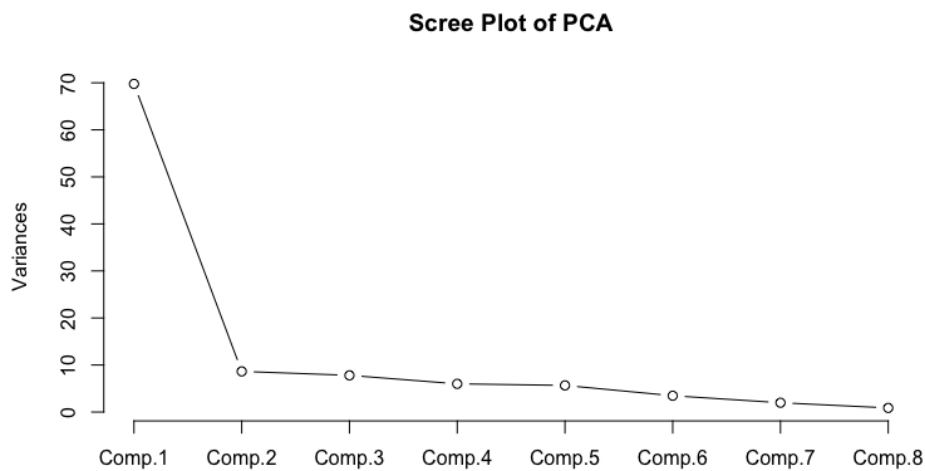


Figure 5: Scree Plot

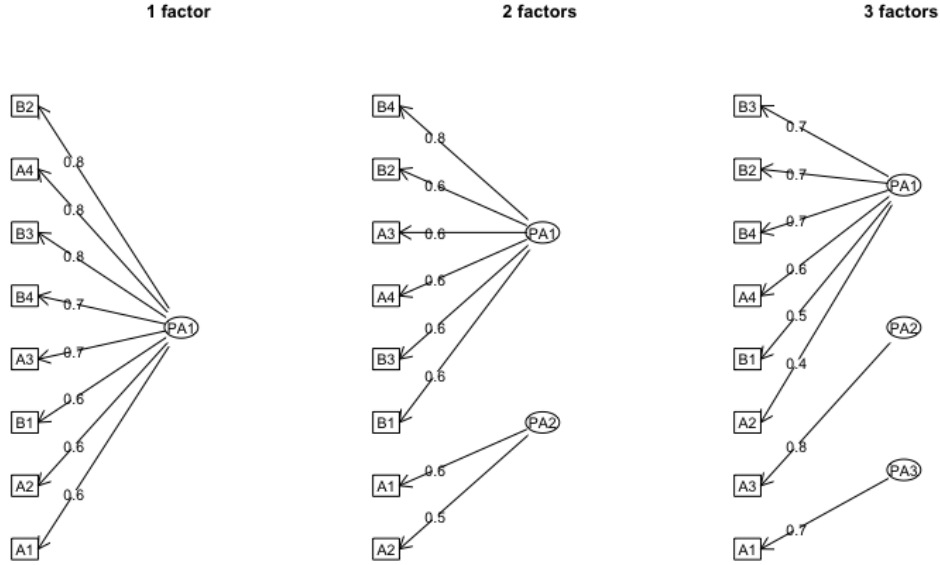


Figure 7: Factor Analysis Diagrams

Factor analysis diagrams (after applying varimax rotation) were produced for three models.

The 1-factor model explains 50% of the variance, with all questions loading onto a single factor, suggesting a broad underlying ability. However, A1 and A2 exhibit high uniqueness, indicating that their variability is not well captured by this model.

The 2-factor model increases explained variance to 54% and introduces a second factor that begins to separate A1 and A2 from the other questions. While this model is statistically acceptable ($p = 0.138$), the second factor contributes only a small amount of additional variance, and the distinction of A1 and A2 remains incomplete.

The 3-factor model provides the best statistical fit ($p = 0.699$) and explains 63% of the variance. It distinctly isolates A1 into a separate factor, while A2 remains associated with the second factor. This suggests that A1 measures a specific skill or dimension not accounted for in simpler models.

4 Particular features of the data

The dataset consists of exam marks for multiple questions, allowing an analysis of student performance across different dimensions. The distribution of scores shows some variability, with certain questions displaying higher uniqueness, particularly A1 and A2, which do not align perfectly with the general performance trend. Principal Component Analysis and Factor Analysis suggest that while most questions contribute to a common underlying ability, some questions measure distinct skills, leading to a more complex structure.

The separation of question groups in the biplot further supports this finding, revealing that specific questions cluster together based on shared properties.

5 Conclusions for the client

The analysis of our exam marks dataset suggested some differences in performances among different questions.

PCA showed that most variance is captured by two principal components: PC1 reflects overall performance, while PC2 differentiates subsets of questions.

Factor analysis supports this, with the 1-factor model failing to explain A1 and A2's variability, while the 2-factor model begins to separate them. The 3-factor model, offering the best statistical fit, isolates A1 as a distinct factor, suggesting it measures a unique skill.

Clustering analysis reinforces these patterns. Single and average linkage clustering reveal that A1 and A2 form a distinct group, indicating they assess a different ability than the other questions.

This supports the idea that student performance is influenced by both general proficiency and specific cognitive skills. Together, Cluster Analysis, PCA and Factor Analysis provide a comprehensive view of the underlying structure of the exam.

Student performance seems not to be driven by a single factor but rather by a combination of general ability and specific skills, with certain questions measuring distinct dimensions of knowledge or cognitive abilities.

6 Appendix

```

1 # Load dataset
2 exammarks <- read.table("exam.txt", header = T)
3 str(exammarks)
4 examcore <- exammarks[-1,-9]
5 n <- nrow(examcore)
6
7 # Load required libraries
8 library(corrplot)
9 library(psych)
10 library(reshape2)
11 library(ggplot2)
12 library(GGally)
13 library(gridExtra)
14
15 # Separate A and B questions
16 marks_A <- examcore[,1:4]
17 marks_B <- examcore[5:8]
18
19 # Standardize data for comparison
20 marks_A_stand <- scale(marks_A)
21 marks_B_stand <- scale(marks_B)
22
23 median_A_stand <- median(unlist(marks_A_stand)) # 0.198
24 median_B_stand <- median(unlist(marks_B_stand)) # 0.2425
25
26 # Adjust margins for boxplots
27 par(mar = c(5, 5, 2, 2))
28 par(mfrow = c(1, 2))
29
30 # Boxplots for standardized marks
31 boxplot(marks_A_stand, main="Amarks", xlab="Aquestions", ylab="Marks", col="
    lightblue")
32 boxplot(marks_B_stand, main="Bmarks", xlab="Bquestions", ylab="Marks", col="
    lightgreen")
33
34 # Parallel Coordinate Plots
35 par(mfrow = c(1, 2))
36 par(mar = c(4, 4, 2, 1))
37 plot1 <- ggparcoord(marks_A_stand)
38 plot2 <- ggparcoord(marks_B_stand)
39 grid.arrange(plot1, plot2, ncol = 2)
40
41 # Convert data to long format for ggplot
42 examcore_long <- melt(as.data.frame(examcore), variable.name = "Question",
    value.name = "Score")
43
44 # Jitter plot of exam scores
45 ggplot(examcore_long, aes(x = Question, y = Score)) +
46   geom_jitter(width = 0.2, alpha = 0.5, color = "blue") +
47   labs(title = "Jitter Plot of Exam Scores", x = "ExamQuestion", y = "Score")
48   +
49   theme_minimal()

```

```

50 ##### CLUSTER ANALYSIS #####
51 par(mfrow = c(1,2))
52 label.x <- c("A1", "A2", "A3", "A4", "B1", "B2", "B3", "B4")
53 unicor <- cor(examcore)
54 cordist <- 0.5-unicor/2
55 dist <- as.dist(cordist)
56
57 ## Complete Linkage ##
58 h <- hclust(d = dist, method = "complete")
59 plot(x = h, labels = label.x, main="Complete Linkage")
60
61 h.cl <- c()
62 h.cl[1] <- h$height[1]
63 for(i in 2:length(h$height)) h.cl[i] <- (h$height[i] - h$height[i-1])
64 which.max(h.cl)
65
66 h.max <- h$height[which.max(h.cl)]
67 abline(h=h.max, col="purple")
68
69 ## Average Linkage ##
70 h1 <- hclust(d = dist, method = "average")
71 plot(x = h1, labels = label.x, main="Average Linkage")
72
73 h.cl <- c()
74 h.cl[1] <- h1$height[1]
75 for(i in 2:length(h1$height)) h.cl[i] <- (h1$height[i] - h1$height[i-1])
76 which.max(h.cl)
77 h.max1 <- h1$height[which.max(h.cl)]
78 abline(h=h.max1, col="green")
79
80 #### PCA ####
81 # Compute correlation matrix
82 R <- as.matrix(cor(examcore))
83 n <- nrow(examcore)
84
85 par(mfrow = c(1,1))
86 corrplot(R, method = "number", type = "upper", tl.cex = 0.8, number.cex = 0.8,
87         col = colorRampPalette(c("white", "salmon", "darkblue"))(200))
88
89 # PCA computation
90 pca_result <- prcomp(examcore, scale = TRUE)
91 summary(pca_result)
92 round(pca_result$rotation, 3)
93
94 # Scree plot for PCA
95 plot(pca_result, type = "lines", main = "Scree Plot of PCA")
96
97 # PCA biplot
98 biplot(pca_result, col = c("blue", "red"), cex = c(0.8, 1.2))
99
100 # Extract rows corresponding to outlier students from exammarks
101 outliers <- exammarks[c("TWFC1", "XDWP6", "VXDZ7", "VNCR6", "SKNR1", "TCCR9", "
102   TXSR2", "WKBC4", "XBVY0", "SXYK4", "TLFB3"), ]
103 outliers$Student <- rownames(outliers)
104 outliers <- outliers[, c("Student", setdiff(colnames(outliers), "Student"))]
105 print(outliers)
106
107 ##### FACTOR ANALYSIS - PFA #####
108 pfa1 <- fa(r = R, nfactors = 1, fm = "pa", rotate = "none")
109 pfa2 <- fa(r = R, nfactors = 2, fm = "pa", rotate = "none")
110 pfa3 <- fa(r = R, nfactors = 3, fm = "pa", rotate = "none")
111
112 # Function to print factor analysis results

```

```

111 print_results <- function(model, nfactors) {
112   cat("\n---", nfactors, "Factor_Model---\n")
113   cat("\nLoadings:\n")
114   print(model$loadings)
115   cat("\nCommunalities:\n")
116   print(model$communality)
117 }
118
119 # Print results for all models
120 print_results(pfa1, 1)
121 print_results(pfa2, 2)
122 print_results(pfa3, 3)
123
124 # Extract communalities for different models
125 comm_1 <- pfa1$communality
126 comm_2 <- pfa2$communality
127 comm_3 <- pfa3$communality
128 communalities <- as.data.frame(cbind(comm_1, comm_2, comm_3))
129 colnames(communalities) <- c("1-Factor_Model", "2-Factor_Model", "3-Factor_
    Model")
130 print(communalities)
131
132 # Varimax rotation for factor analysis
133 pfa1varimax <- fa(r = R, nfactors = 1, fm = "pa", rotate = "varimax")
134 pfa2varimax <- fa(r = R, nfactors = 2, fm = "pa", rotate = "varimax")
135 pfa3varimax <- fa(r = R, nfactors = 3, fm = "pa", rotate = "varimax")
136
137 pfa1varimax$loadings
138 pfa2varimax$loadings
139 pfa3varimax$loadings
140
141 # FA diagrams
142 par(mfrow = c(1,3))
143 fa.diagram(pfa1varimax, main = "1_factor")
144 fa.diagram(pfa2varimax, main = "2_factors")
145 fa.diagram(pfa3varimax, main = "3_factors")
146
147 # Maximum Likelihood Factor Analysis (MLFA)
148 mlfa1 <- factanal(covmat = R, factors = 1, n.obs = n, rotation="none")
149 mlfa2 <- factanal(covmat = R, factors = 2, n.obs = n, rotation="none")
150 mlfa3 <- factanal(covmat = R, factors = 3, n.obs = n, rotation="none")
151
152 # Print MLFA results
153 print(mlfa1, digits = 3)
154 print(mlfa2, digits = 3)
155 print(mlfa3, digits = 3)

```

Listing 1: R Code for Exam Marks Analysis